Lab 3: Light-Emitting Diodes

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ESE 2180: Electronic, Photonic and Electromechanical Devices

In this lab, you will study the properties of light-emitting diodes (LEDs). As you already know, LEDs are p-n junctions fabricated from various types of doped direct band gap semiconductors (such as III-V and II-VI). When operated in the forward bias, radiative recombination at the depletion region releases photons of light with energies equal to the bandgap energy in what is known as electroluminescence (production of light due to the flow of current).

The bandgaps of these semiconductors correspond to energies of light in the visible spectrum as per Planck's relation,

$$E_g = h\nu,$$

where h is Planck's constant and ν is the frequency of light. We can tune the color of light the LED emits by choosing semiconductors of desired bandgaps, or by growing epitaxially various materials on top of each other to engineer our desired bandgap. For the latter process, we need to take into consideration the lattice constants of different materials, as better lattice constant matching reduces the possibilities of defects being induced in our device.

Section A explores the electronic properties of LEDs stemming from their I-V characteristics, while Section B explores the optical properties of LEDs stemming from their semiconductor compositions.

Part A: Current Voltage Characteristics of a (Light Emitting) Diode

You are given three LEDs (red, green, and white). First, you will measure the current voltage (I-V) characteristics of the diode.

The following steps should be repeated for three LEDs, one at a time:

1. Connect the LED, mounted on a plastic adaptor, to the power supply and ammeter (in series). The current-limiting resistor soldered in series to the LED should prevent the LED from burning out.
2. Start at 1 V and ramp up the voltage up to 3.5 V in increments of 0.1 V. Record the current value at each voltage. Also, note down the voltage at which you first see the LED light up.

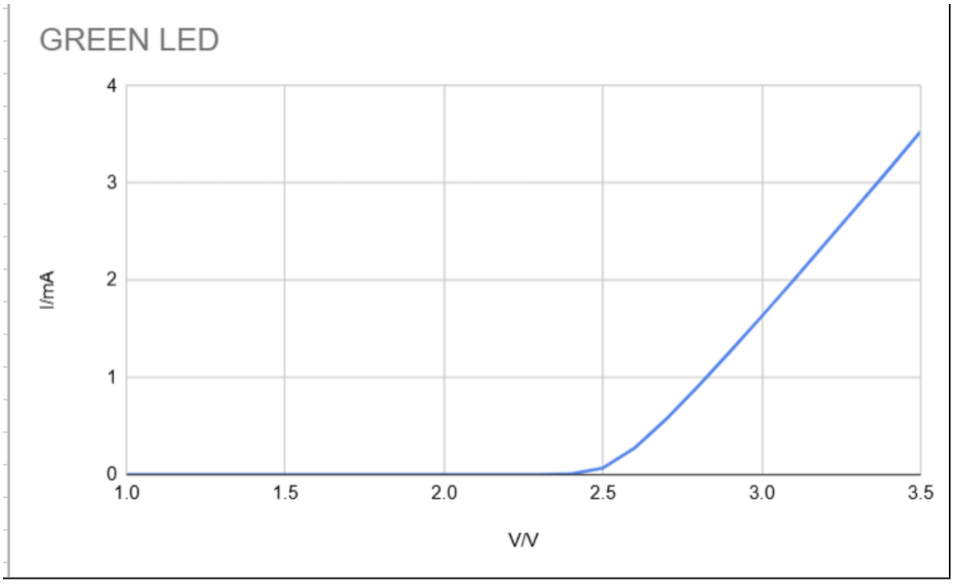
LEDs are diodes, meaning that they pass respectable current only in forward bias. Make sure you are connecting your diodes properly. If you are seeing no light at 2.5 V, you have not connected properly and may want to flip the wires. **Do not increase voltage beyond** 3.5 V **for any LED.**

This is a discontinuous I-V data set and will produce a jagged I-V curve. Because the voltage is sampled in 0.1 V increments, this measured data set has a lower voltage resolution than the continuous data set provided for analysis. Acquiring a continuous data set (i.e. ramping up the voltage by extremely small increments and measuring the current) would require complex and expensive electrical measurement equipment. We provide you this data set for each LED. In the data processing section, you will be handling both the discontinuous and the continuous data sets.

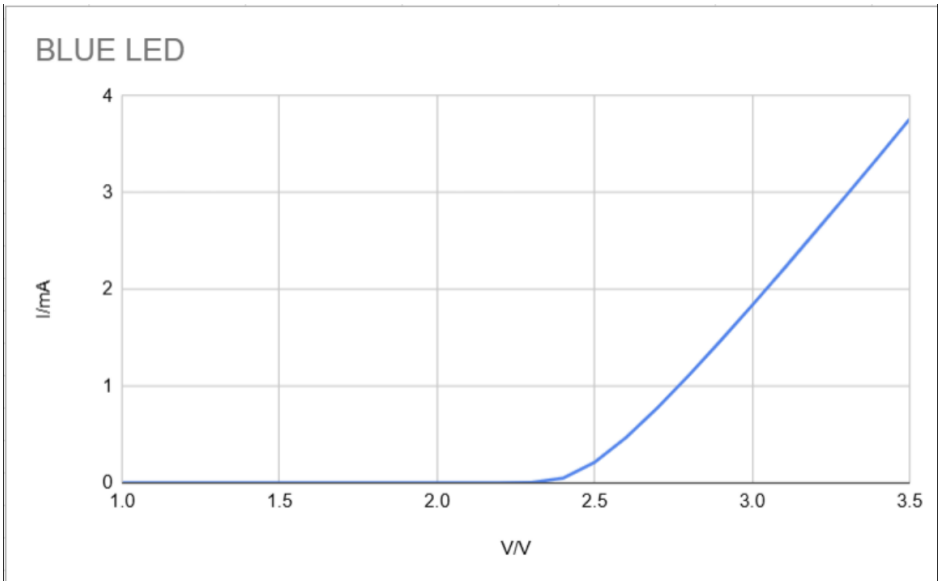
Part A: Data Processing, Discussion, and Analysis

Answer the following questions with short responses, max 1–2 sentences. Show calculations where requested.

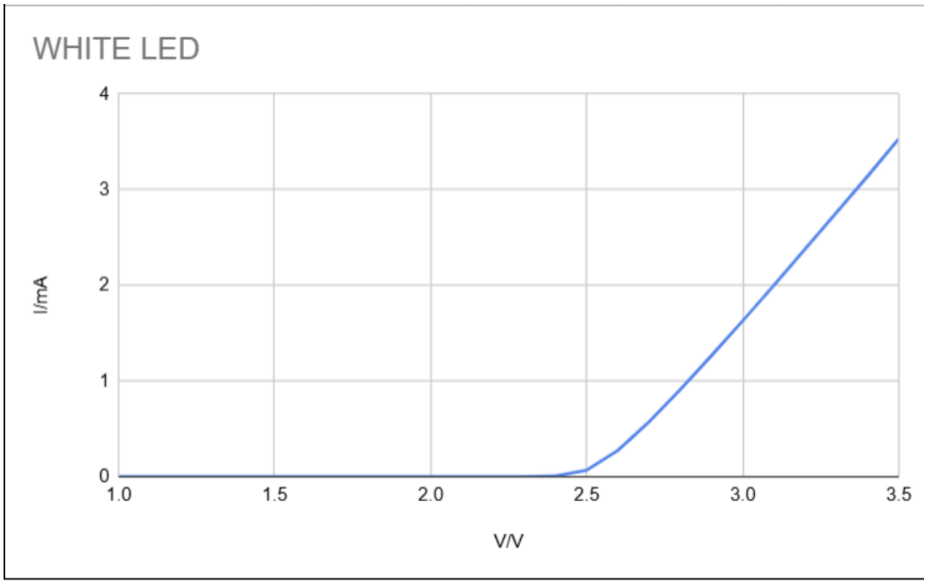
(a) Using the discrete data set you measured, plot the I-V characteristics for each of the LEDs and estimate the forward voltage. Extrapolate the voltage at which the I-V curve begins to noticeably increase to find the forward voltages and record them below. **(0.75 pt)**



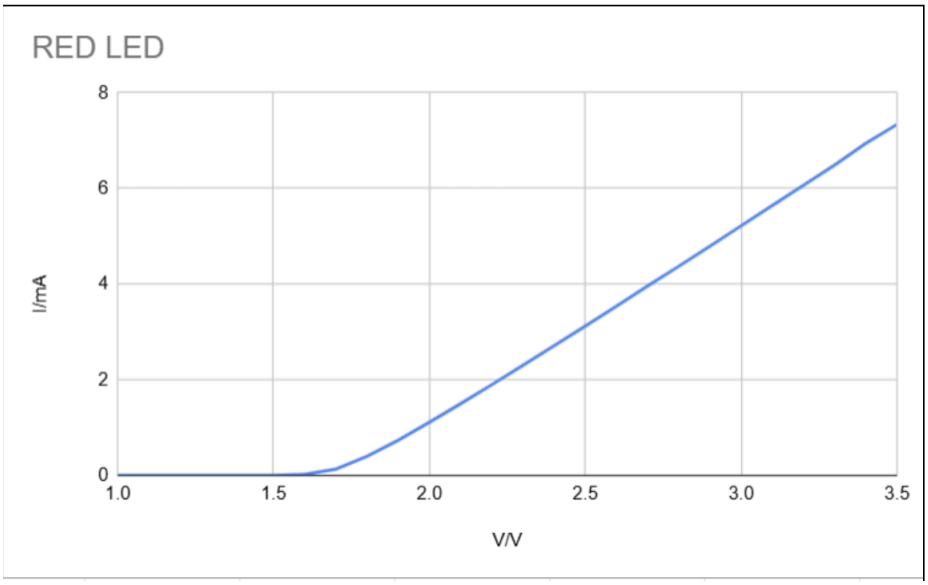
Forward voltage approximately 2.5 V.



Forward voltage approximately 2.4 V. We expected this to be the highest as it has the largest band gap, but our data showed otherwise. This may be due to error.



Forward voltage approximately 2.5 V.



Forward voltage approximately 1.7 V.

(b) Using the continuous data set provided, plot the I-V characteristics for the red LED. Consider the I-V characteristic of a p-n junction diode. Explain the following features: **(0.75 pt)**

(i) **Why is the current approximately $-I_0$ under reverse bias?**
Under reverse bias, majority-carrier flow is suppressed and so the current is dominated by thermally generated minority carriers that are pulled through the depletion region. Their concentration is independent of the applied reverse voltage, so the current approaches the constant reverse saturation current $-I_0$.

(ii) **Why is the net current zero at $V = 0$?**
At $V = 0$, the diode is in thermal equilibrium, so the diffusion current caused by carrier concentration gradients is balanced by the drift current from the built-in electric field. You can also see this through the equations.

(iii) **Why does the current increase exponentially with applied voltage under forward bias?**
Forward bias will lower the potential barrier of the junction, leading to the injected minority-carrier concentration increasing exponentially with voltage. And so,

$$I = I_0 e^{qV/(nkT)}.$$

(c) Using the continuous data set provided, plot the I-V characteristics for each of the LEDs and estimate the forward voltage. Extrapolate the voltage at which the I-V curve begins to noticeably increase to find the forward voltage and record them below. **(0.75 pt)**

LED	V_F continuous
Red	1.7 V
Green	2.3 V
Blue	2.4 V
White	2.6 V

(d) Replot the continuous data set (not the measured discrete data set) as log(current) vs. voltage. Identify the voltage range over which the curve is approximately linear and report this range. **(0.5 pt)**

LED	Approximate linear voltage range
Red	1.58–1.76 V
Green	2.00–2.14 V
Blue	2.28–2.46 V
White	2.34–2.56 V

(e) Consider the diode equation to fit your I-V characteristics in the forward bias direction, noting

$$I = I_0 \left[\exp\left(\frac{qV}{kT}\right) - 1 \right] \approx I_0 \exp\left(\frac{qV}{kT}\right)$$

in forward bias. Most diodes are very close to this ideal equation in the forward bias, but not always. The more general equation is

$$I = I_0 \left[\exp\left(\frac{qV}{nkT}\right) - 1 \right] \approx I_0 \exp\left(\frac{qV}{nkT}\right),$$

where n describes the non-ideality. Most diodes have ideality factor ranging from 1 to 2. Fit the log(current) vs. voltage curves for your diodes (from the continuous data set) to this more general equation and for each color solve for n. Tabulate these n values below. **(0.75 pt)**

LED color	V_F (discrete)	V_F (continuous)	Voltage range	$n = \frac{38.68}{\text{slope}}$
Red	1.7 V	1.7 V	1.58–1.76 V	1.46
Green	2.5 V	2.3 V	2.00–2.14 V	1.88
Blue	2.4 V	2.4 V	2.28–2.46 V	1.80
White	2.5 V	2.6 V	2.34–2.56 V	1.33

We used $T = 300\,\text{K}$.

(f) Compare the forward voltages obtained from the discrete and continuous data sets. Which data set should provide a more precise estimate of the forward voltage, and why? You do not need to look up literature values for these voltages to corroborate your claim. **(0.5 pt)**

The continuous dataset will provide a much more precise estimate of the forward voltage because it samples the I-V characteristic at much smaller voltage increments (the limit as the discrete increments become very small). The discrete measurements use 0.1 V steps, so the location of the diode's turn-on region can only be determined to roughly that scale.

Part B: LED Color, Wavelength and Composition

You should come down to the basement of the Singh Center!

In this part of the lab, a constant voltage is applied to four LEDs (red, green, blue, and white), and the intensity of the emitted light is measured over a range of wavelengths.

1. Connect each LED (red, green, blue, and white) in series with a 220 Ω resistor. Apply a forward-bias voltage of 3.5 V.
2. If the measured spectrum is saturated (i.e. you observe a plateau at the peak), reduce the integration time accordingly.
3. Measure the peak wavelength for each LED and take a picture of each spectrum with your cell phone.
4. Point the red laser at the detector and compare its spectrum with that of the red LED.
5. Point your phone flashlight at the detector and compare its spectrum with that of the white LED.

Part B: Data Processing, Discussion, and Analysis

(a) Measure peak wavelength for Green, Red, Blue, and White LEDs with fluorescence spectroscopy. From these plots, identify the band gap energy of each LED. White may have 3 or more; estimate band gaps associated with each peak. **(2 pts)**

LED color	Peak 1 wavelength (nm)	Peak 2 wavelength (nm)	Peak 3 wavelength (nm)
Green	528.57	N/A	N/A
Red	631.32	N/A	N/A
Blue	472.55	N/A	N/A
White	450.52	532.35	555.19

LED color	Peak 1 bandgap (eV)	Peak 2 bandgap (eV)	Peak 3 bandgap (eV)
Green	2.346	N/A	N/A
Red	1.964	N/A	N/A
Blue	2.624	N/A	N/A
White	2.752	2.329	2.233

(b) Determine the composition of the colored LEDs using the plots attached in the next page. For the Blue and Green LED, determine x in $In_xGa_{1-x}N$. If you noted two peaks for Green, determine x for both. For the Red LED, determine x in $Al_xGa_{1-x}As$ and x in $In_xGa_{1-x}P$. **(2 pts)**

Blue LED:

$$E_g = 2.624\,\text{eV} \quad \Longrightarrow \quad x = 0.20,$$

so the composition is

$$In_{0.20}Ga_{0.80}N.$$

Green LED:

$$E_g = 2.346\,\text{eV} \quad \Longrightarrow \quad x = 0.28,$$

so the composition is

$$In_{0.28}Ga_{0.72}N.$$

Red LED:

$$E_g = 1.964\,\text{eV} \quad \Longrightarrow \quad x = 0.46,$$

so the composition is

$$In_{0.46}Ga_{0.54}P.$$

For AlGaAs,

$$x = \frac{1.964 - 1.422}{1.2475} = 0.434,$$

so the composition is approximately

$$Al_{0.43}Ga_{0.57}As.$$

(c) Do a little research on the fabrication of $Al_xGa_{1-x}As$ and $In_xGa_{1-x}P$. Which one would you consider a more ideal choice for fabricating Red LEDs? **(0.5 pts)**

$In_xGa_{1-x}P$ is the better choice for a red LED since a red-emitting AlGaAs composition lies close to its direct-to-indirect bandgap crossover, which reduces the efficiency of radiative recombination. InGaP can provide a direct bandgap near the desired red photon energy while also allowing good lattice matching to GaAs.

(d) Discuss the difference you saw between the red LED and the red laser spectra. In what ways do they differ and why? **(0.5 pts)**

Between the red LED vs. the red laser spectra, they both peaked at the same wavelength; however, the laser's peak was much sharper and more distinct. The LED emitted over a broader range of wavelengths.

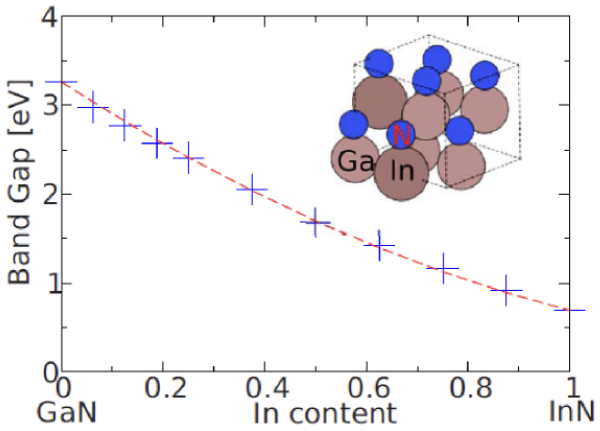
(e) For the white LED, what colors/energy does each peak correspond to? Why are some peaks higher than others? What would be the likely compositions based on the plots on the next page? **(0.5 pts)**

The 450.52 nm peak corresponds to blue light, the 532.35 nm peak to green light, and the 555.19 nm peak to yellow-green light. Their different heights come because the white LED does not emit equal intensity for every wavelength; the relative intensities depend on the wavelength. From the supplied InGaN plot, the peaks will approximately correspond to $x = 0.17$, 0.29, and 0.32, respectively.

(f) Discuss the difference you saw between the white LED and the phone flashlight spectra. In what ways do they differ and why? **(0.5 pts)**

The iPhone flashlight is a little more complex than the white LED as it consists of a much broader spectrum of wavelengths. It was much harder to see distinct peaks, as there were more than 5 peaks as opposed to 3 that interfered with one another, making it difficult to determine the exact wavelengths at the peaks.

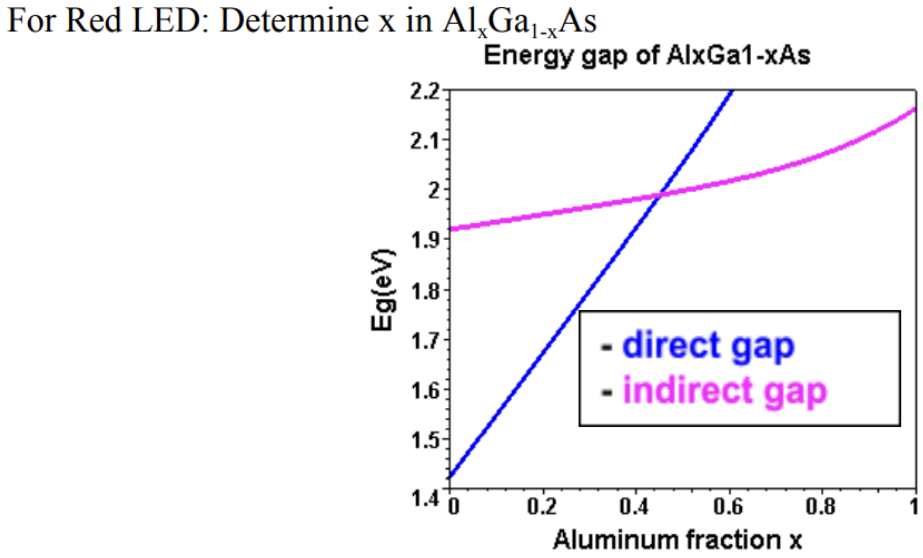
Reference Plots



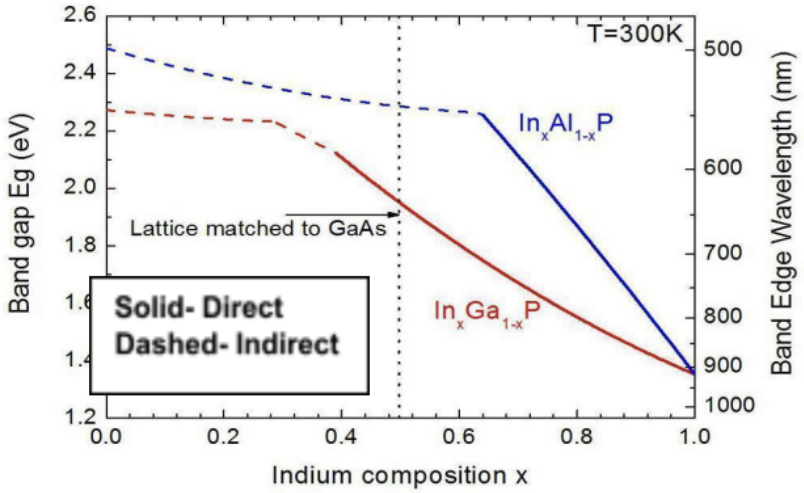
f: Appl. Phys. Lett. 96, 021908 (2010)

Reference: *Applied Physics Letters* **96**, 021908 (2010).

For Red LED: Determine x in $Al_xGa_{1-x}As$



Source: `https://www.batop.de/information/Eg_AlGaAs.html`



Reference: Zhang and Gu, `http://dx.doi.org/10.5772/50404`